

The Flexible Global Ocean-Atmosphere-Land System Model, Grid-point Version 2: FGOALS-g2

LI Lijuan¹ (李立娟), LIN Pengfei¹ (林鹏飞), YU Yongqiang¹ (俞永强), WANG Bin^{*1,2} (王 斌),
ZHOU Tianjun¹ (周天军), LIU Li² (刘 利), LIU Jiping¹ (刘骥平), BAO Qing¹ (包 庆),
XU Shiming² (徐世明), HUANG Wenyu² (黄文誉), XIA Kun² (夏 坤), PU Ye¹ (普 业),
DONG Li¹ (董 理), SHEN Si¹ (申 思), LIU Yimin¹ (刘屹岷), HU Ning¹ (胡 宁), LIU Mimi¹ (刘咪咪),
SUN Wenqi¹ (孙文奇), SHI Xiangjun⁴ (史湘军), ZHENG Weipeng¹ (郑伟鹏), WU Bo¹ (吴 波),
SONG Mirong¹ (宋米荣), LIU Hailong¹ (刘海龙), ZHANG Xuehong¹ (张学洪), WU Guoxiong¹ (吴国雄),
XUE Wei² (薛 巍), HUANG Xiaomeng² (黄小猛), YANG Guangwen² (杨广文),
SONG Zhenya³ (宋振亚), and QIAO Fangli³ (乔方利)

¹*State Key Laboratory of Numerical Modeling for Atmospheric Sciences and Geophysical Fluid Dynamics,
Institute of Atmospheric Physics, Chinese Academy of Sciences, Beijing 100029*

²*Ministry of Education Key Laboratory for Earth System Modeling, Center of Earth System Science,
Tsinghua University, Beijing 100084*

³*First Institute of Oceanography, State Oceanic Administration, Qingdao 266061*

⁴*Climate Center, Hebei Meteorological Administration, Shijiazhuang 050021*

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ABSTRACT

This study mainly introduces the development of the Flexible Global Ocean-Atmosphere-Land System Model: Grid-point Version 2 (FGOALS-g2) and the preliminary evaluations of its performances based on results from the pre-industrial control run and four members of historical runs according to the fifth phase of the Coupled Model Intercomparison Project (CMIP5) experiment design. The results suggest that many obvious improvements have been achieved by the FGOALS-g2 compared with the previous version, FGOALS-g1, including its climatological mean states, climate variability, and 20th century surface temperature evolution. For example, FGOALS-g2 better simulates the frequency of tropical land precipitation, East Asian Monsoon precipitation and its seasonal cycle, MJO and ENSO, which are closely related to the updated cumulus parameterization scheme, as well as the alleviation of uncertainties in some key parameters in shallow and deep convection schemes, cloud fraction, cloud macro/microphysical processes and the boundary layer scheme in its atmospheric model. The annual cycle of sea surface temperature along the equator in the Pacific is significantly improved in the new version. The sea ice salinity simulation is one of the unique characteristics of FGOALS-g2, although it is somehow inconsistent with empirical observations in the Antarctic.

Key words: FGOALS-g2, climatological mean state, climate variability, 20th century climate, monsoon

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1. Introduction

Climate system models (CSMs) play an instrumental role in understanding and simulating past, present,

and future climates. In recent decades, great effort has been made developing component models and fully coupled models at the State Key Laboratory of Numerical Modeling for Atmospheric Sciences and Geo-

*Corresponding author: WANG Bin, wab@lasg.iap.ac.cn

physical Fluid Dynamics (LASG), the Institute of Atmospheric Physics (IAP) of the Chinese Academy of Sciences. The early achievements of the LASG/IAP in the development and application of CSMs from the end of the 1980s to the year 2007 are reviewed in Zhang et al. (2000), Yu and Zhou (2004), Zhou et al. (2007) and Wang et al. (2009a), along with systematic summaries of climate processes, variability and climate change studies performed with these models.

The Flexible Global Ocean-Atmosphere-Land System model (FGOALS) is composed of separate component models of the atmosphere, ocean, sea ice, and land surface. The preliminary version of FGOALS extends back to the early 2000s, as an achievement of the LASG/IAP's decision to consolidate and modernize its numerical modeling activities through the development of a new coupled system employing a coupler structure (Yu et al., 2002). One of the preliminary versions, referred to as FGOALS-g1.0 (Yu et al., 2008), was developed in 2005 by employing a new atmospheric component, i.e. the Grid-point Atmospheric Model of IAP LASG, version 1 (GAMIL1) (Wang et al., 2004; Li et al., 2007a, b; Li and Wang, 2010). A twin version of FGOALS-g1.0 is FGOALS-s1.0, which employs the Spectral Atmospheric Model of the IAP/LASG (SAMIL) as its atmospheric component (Zhou et al., 2005). To reduce the need for computing power and resources in large ensemble or millennial-scale climate simulations, a fast coupled version of FGOALS-g was also developed. It shares the same physical and dynamical processes of the standard version of FGOALS-g1.0, but uses reduced resolution and increased time-step (Zhou et al., 2008). By combining different versions together, the LASG/IAP aims to create a common flexible modeling infrastructure, which balances scientific needs with resource availability. The FGOALS system supports diverse research activities, from seasonal monsoon evolution to anthropogenic climate change, and from millennial climate evolution to paleoclimate simulation.

FGOALS Grid-point version 2 (FGOALS-g2) is one of the newest versions of FGOALS. It has been developed by a community of scientists and graduate students from the LASG/IAP, Center for Earth System Science (CESS)/Tsinghua University, and the First Institute of Oceanography (FIO)/State Oceanic Administration. Its previous version, FGOALS-g1.0, was one of many climate models around the world that participated in the 3rd phase of the Coupled Model Intercomparison Project (CMIP3), the results of which were cited by the 4th Assessment Report (AR4) of the IPCC. The FGOALS-g1.0 model produces a realistic simulation of climate in many respects, as described in Zhou et al. (2007), Yu et al. (2008) and Zeng et al.

(2012). However, this model exhibits a large bias at high latitudes, for example a cold bias in surface temperature and weaker Atlantic Meridional Overturning Circulation (AMOC) and meridional heat transport. Reducing the mid-latitude climate drifts was the highest priority when developing FGOALS-g2, for which a significant improvement has been achieved. In fact, these model biases were alleviated in an earlier intermediate version of the model, FGOALS-g1.1, through improving the air-sea coupling scheme and dynamical framework of the ocean component model (Yu et al., 2011). FGOALS-g2 is one of two versions of the FGOALS model being finalized for participation in CMIP5. The CMIP5 protocol calls for both "Long-Term" (century and longer) and "Near-Term (decadal prediction)" simulations. Results from these runs using FGOALS-g2 have been submitted to CMIP5 for inclusion in the 5th Assessment Report (AR5) of the IPCC via an Earth System Grid data node (ESG-Node) constructed at the LASG/IAP. This overview paper describes some of the most important developments and improvements in the model's components, as well as the model's performance compared with the previous version, FGOALS-g1.0.

The remainder of the paper is organized as follows. Section 2 documents the major developments in FGOALS-g2 components since the previous version. The stability of the coupled system and the spin-up processes are also described here. Section 3 shows results from climate mean states to major climate variability modes. The performances of 20th century climate and Asian-African Monsoon simulations are also assessed here. Finally, section 4 provides a summary and discussion.

2. Model description and experiment design

2.1 Model description

The four components of FGOALS-g2 include: the atmospheric component (GAMIL2); the oceanic model (LICOM2; LASG IAP climate system ocean model version 2); the sea ice component (CICE4-LASG; Community Ice Code), an improved version of CICE4.0 (Los Alamos sea ice model version 4.0, <http://climate.lanl.gov/Models/CICE>); and the introduced Community Land Model (CLM3) from the National Center for Atmospheric Research (NCAR). All the model components are connected by the coupler CPL6 (Craig et al., 2005) from the NCAR.

GAMIL2 employs a hybrid horizontal grid, with a Gaussian grid of 2.8° between 65.58°S and 65.58°N and a weighted equal-area grid poleward of 65.58° .

There are 26 vertical σ -layers (pressure normalized by surface pressure) with the model top at 2.194

hPa. Its dynamical core includes a finite difference scheme that conserves mass and effective energy while solving the primitive hydrostatic equations of baroclinic atmosphere (Wang et al., 2004), and a two-step shape-preserving advection scheme (TSPAS) for the moisture equation (Yu, 1994). Besides the same dynamical core and grid as in GAMIL1, GAMIL2 also adopts the same cloud fraction scheme (Slingo, 1987; Collins et al., 2003). The main differences between the two versions, GAMIL1 (Wang et al., 2004; Li et al., 2007a, b; Li and Wang, 2010) and GAMIL2 (Li et al., 2012), are the upgraded cloud-related processes, for example the deep convective parameterizations (Zhang and McFarlane, 1995; Zhang and Mu, 2005a), convective cloud fraction (Xu and Krueger, 1991; Rasch and Kristjánsson, 1998) and microphysical schemes (Rasch and Kristjánsson, 1998; Morrison and Gettelman, 2008; Shi et al., 2010). Besides, for the energy balance at top of atmosphere (TOA), some parameters in convection (including shallow and deep convection), cloud macro/micro-physical and boundary layer schemes have changed in GAMIL2, the detail of which can be found in Li et al. (2013). In addition, the large impact of the parameter changes on cloud simulations has been explored using the CAPT (the Climate Change Prediction Program and Atmospheric Radiation Measurement Program: the CCM-ARM Parameterization Testbed) method and the COSP (Cloud Feedback Model Intercomparison Project (CFMIP) Observation Simulator Package; see Bodas-Salcedo et al., 2011) (Dong et al., 2012; Li et al., 2012).

LICOM is the fourth generation of the LASG Ocean General Circulation Model (OGCM) (Liu et al., 2004a, b), the first generation of which was developed in 1989 (Zhang and Liang, 1989). Version 2 (LICOM2) is used in FGOALS-g2. It adopts the geographic longitude–latitude grid and η -vertical coordinates (30 levels). For model stability, the North Pole is simply treated as land. Compared with LICOM1.0 and LICOM1.1, LICOM2 features increased horizontal resolutions ($1^\circ \times 1^\circ$ horizontal resolution with 0.5° meridional resolution in the tropics) and an adjusted vertical resolution in the upper layers (10 m for each layer in the upper 150 m) (Wu et al., 2005). In addition, the TSPAS advection scheme has been introduced (Xiao, 2006); plus, the physical processes have been updated or improved (Liu et al., 2012), including the mixing schemes (Canuto et al., 2001, 2002; Liu et al., 2012), solar penetration scheme (Lin et al., 2007, 2011), and other physical processes (Liu et al., 2012). Several updated schemes have not been adopted in the latest versions of FGOALS (Lin et al., 2013a), the details of which can be found in Liu et al. (2012) and Lin et al. (2013a).

CICE4-LASG is an improved version of CICE4.0. One of the main differences between CICE4-LASG and CICE4.0 is the improvement of sea ice salinity. Sea ice salinity is characterized by large spatiotemporal variations. This affects sea ice thermal properties and flux exchanges between sea ice and ocean, which strongly influence sea ice mass balance, oceanic density stratification and thermohaline circulation. However, CICE4.0 uses the prescribed and time-invariant vertical salinity profiles that misrepresent ice salinity. To address this issue, a simple parameterization for brine entrapment and drainage was designed and applied in CICE4-LASG (Wang et al., 2009b; Liu, 2010).

CLM3 (Oleson et al., 2010), introduced to the coupled model FGOALS-g2, is based on the Biosphere–Atmosphere Transfer Scheme (BATS), the IAP Land Surface Model (IAP94), and the Land Surface Model (LSM). The NCAR’s Community Climate System Model (CCSM) coupler CPL6 (Craig et al., 2005) is used to connect all the component models in FGOALS-g2 and control the parallel integration of the whole coupled model. The updating frequency of the component models is once per hour, except for the oceanic component which is once per day. All of the experiments are performed on the supercomputer Tansuo 100 in the Tsinghua National Laboratory for Information Science and Technology, with about 30 model years run per day using 108 CPU cores.

2.2 Experiment design

This study focuses on the monthly and daily mean outputs from the pre-industrial (PI) control runs and historical runs according to the experiment design of CMIP5 (Taylor et al., 2009). The values of external forcing have been kept to the year 1850 in all of the PI control runs, in which, the solar constant is 1365 W m^{-2} and the greenhouse gas mixing ratios (including CO_2 , CH_4 and N_2O) are 284 ppm, 790 ppb and 275 ppb, respectively. The ozone mixing ratio and the concentrations of aerosols including sulphate, black and organic carbon, dust, and sea salt are prescribed as the values in the PI period. The PI control run with more than 1000 years of integrations starts from the equilibrium state of the stand-alone OGCM after a 500-yr spin-up with LICOM2. It produces a $-0.02^\circ\text{C} (100 \text{ yr})^{-1}$ ($-0.011 \text{ \% yr}^{-1}$) trend (Lin et al., 2013c) in annual global SST, and $5.87 \times 10^3 \text{ km}^2 (100 \text{ yr})^{-1}$ ($0.0052 \text{ \% yr}^{-1}$) and $2.20 \times 10^5 \text{ km}^2 (100 \text{ yr})^{-1}$ ($0.13 \text{ \% yr}^{-1}$) trends in Northern and Southern Hemispheric sea ice extent, respectively (Figure not shown). This means there are very small drifts in global SST and Northern Hemispheric sea ice, but relatively obvious drifts in Southern Hemispheric sea ice in the PI control

run (Lin et al., 2013c).

Four members of 20th century runs, forced by the time-varying solar constant, greenhouse gases (GHGs), aerosols and ozone, were completed. Here it should be noted that the volcanic forcing was not included. The solar constant is the annually resolved total solar irradiation (TSI) with the background recommended by CMIP5 (Lean, 2009). The greenhouse gas concentrations are as specified on the IIASA (International Institute for Applied Systems Analysis) website (<http://www.iiasa.ac.at/webapps/tnt/RcpDb/dsd?Action=htmlpage&page=welcome>). The AC&C (Atmospheric Chemistry & Climate)/SPARC (Stratospheric Processes And their Role in Climate) ozone database is used for ozone forcing, which consists of separate stratospheric and tropospheric data sources (Cionni et al., 2011). The aerosol files are monthly 10-yr averages centered around the middle of each decade produced by the NCAR Community Atmospheric Model version 3.5 (CAM3.5) with a bulk aerosol model recommended by CMIP5, and linearly interpolated in time from year to year within each month, which plays a smoothing role. The initial conditions are from years 440, 450, 500, and 520 of the PI control run. All of the uncertainty parameters of the atmospheric and oceanic model are unchanging in all of the CMIP5 experiment runs.

3. Results

3.1 Climatological mean state

3.1.1 Sea surface temperature

SST is one of the most important variables reflecting the performance of the coupled model. The observed distribution of SST is captured well by the four-member ensemble of 20th century runs, such as the warm pool in the western Pacific and Indian Ocean, cold tongue in the east etc. (Fig. 1). From a global perspective, the simulated global mean SST is 0.74°C lower than the observed, and their RMSE is 1.56°C (Fig. 1c). The SST bias has obvious regional features. There are large cold biases (>4°C) along the paths and extensions of western boundary currents (Kuroshio, Gulf Stream and Brazil current south of 30°S), and the Barents Sea, while along the eastern coasts (Pacific and Atlantic) and in the Labrador Sea large warm biases (>4°C) dominate. In addition, in the central South and East Pacific, there is also warm bias, which implies there is an obvious double ITCZ.

Figure 2 shows the mean SST along the equator (2°S–2°N) and its anomaly annual cycles. As in Fig. 1c, there are cold biases of about 1°C in the central and western Pacific, and 1°C or higher warm bi-

ases in the eastern region (Fig. 2a). The observed SST anomaly in the equatorial eastern Pacific shows obvious annual variation, being warmest in spring and coldest in autumn (Fig. 2c). The phase, the seasonal transition and westward propagation of the annual cycle in the eastern Pacific are well presented by FGOALS-g2 according to the observed (Fig. 2b). The annual cycle in FGOALS-g2 is an important improvement compared with the previous version of FGOALS-g (Lin et al., 2011; Yu et al., 2011). In fact, the annual cycles of SST in the eastern Pacific are difficult to simulate well in most non-adjusted coupled models (Mehchoso et al., 1995; Latif et al., 2001). This may be related to the higher resolutions, with a 0.5° meridional resolution in the tropics and 10 m for each layer in the upper 150 m, as well as the updated mixing scheme in its ocean model and the improvement of the cumulus convection scheme in its atmospheric model. The former two aspects contribute to the better climatological mean simulations in the upper ocean and the latter makes for the more reasonable annual cycles of solar radiation reaching the sea surface in the tropical regions. Further mechanistic analysis is needed to uncover the reason for this significant improvement in another study.

3.1.2 Ocean meridional overturning circulation

Figure 3 shows the global and Atlantic meridional overturning circulation (GMOC and AMOC). The overall patterns and positions of cells, water masses and overturning are similar to the calculated values in observations (Lumpkin and Speer, 2007). In the upper layer (~500 m), there are the tropics–subtropics cells, which are mainly driven by wind stress. Between 500–2500 m north of 35°S, there is North Atlantic Deep Water (NADW). Below 2500 m, the Antarctic Bottom Water (AABW) is mainly located north of 50°S and its maximal value (~16 Sv) is located between 30°–20°S at about 3500 m, larger than the observed mainly due to convection in the deep sea and partially to polar overflow from the Antarctic. Around 40°–50°S, the Deacon cell is from the surface to about 4000 m and its maximal value is 28 Sv. Compared with the observation, its location is about 5° equatorward, related to the peak zonal wind stress bias (figure not shown). In high latitudes, polar meridional overturning circulations (MOCs) exist in the Southern and Northern Hemispheres and their values are about 4 Sv and 2 Sv, respectively. The NADW is mainly produced in the Atlantic due to deep convection. Its maximal value (~29 Sv) is located at 35°N between 800–1000 m. Specifically, its value is 23.4 Sv at 26.5°N, which is larger than the observed value (18.5 Sv) at 26.5°N (RAPID; Cunningham et al., 2007). The maximal en-

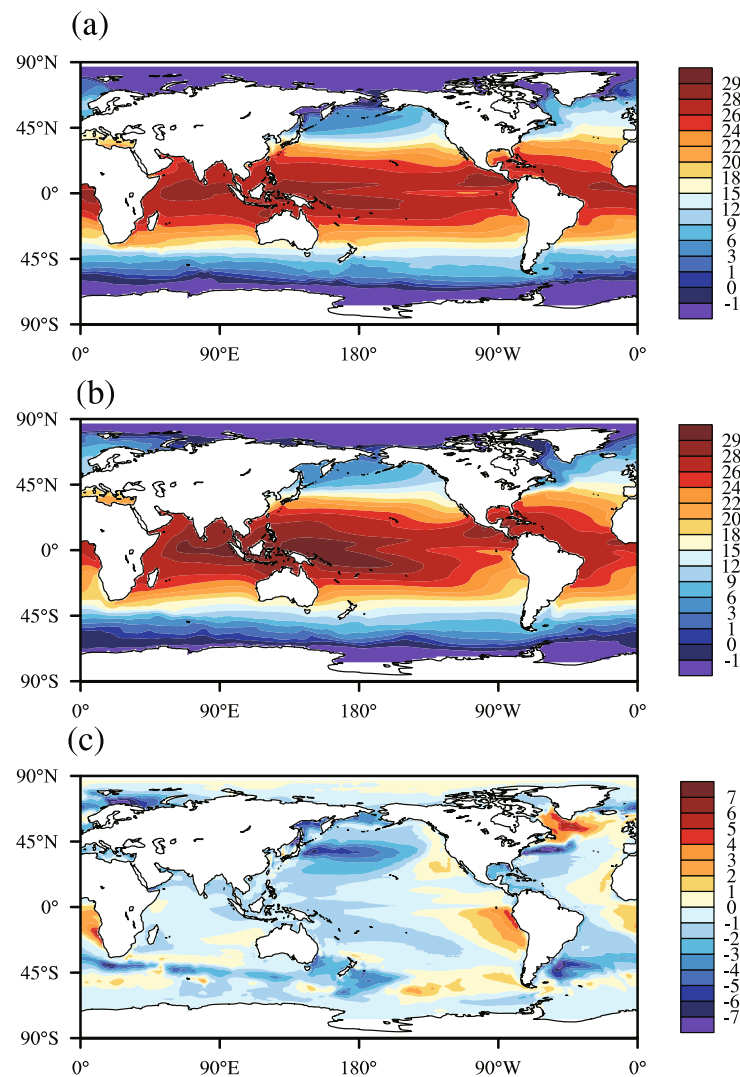


Fig. 1. Climatological mean SST from FGOALS-g2 20th century run (a), observation (b) (Hurrell et al., 2008), and their differences (c) (simulation minus observation) (units: °C).

semble mean AMOC from historical runs is smaller than that from the PI control run (Lin et al., 2013b). For instance, at 26.5°N, the maximal value is reduced from 23.4 Sv in the PI control run to 22.5 Sv in historical runs. The latter is closer to the observed value (18.5 Sv; Cunningham et al., 2007). The AMOC in the historical runs is located within the simulation values in the CMIP3 runs (IPCC, 2007a). Generally, the simulated GMOC and AMOC are significantly improved compared with simulations by the previous version of the model, FGOALS-g1.0 (e.g. Schneider et al., 2007, Lin et al., 2013b). However, the AMOC in FGOALS-g2 is still stronger than its observed value owing to the strong convection in the North Atlantic Ocean, the

detailed reasons and associated mechanisms for which are discussed in Huang et al. (2013)^a.

3.1.3 Sea ice concentration and salinity

Figure 4 shows the spatial distributions of annual average sea ice concentration for the PI control run of FGOALS-g2 for years 541–590. FGOALS-g2 captures well the observed sea ice distributions (Special Sensor Microwave/Imager (SSM/I) data averaged from 1981–2000), both in the Arctic and Antarctic. For the Arctic, the regions with largest ice concentrations simulated by the model have a good spatial correspondence to the observation. However, we note that the model produces much more ice in the Barents Sea, and

^aHuang, W. Y., and Coauthors, 2013: Variability of Atlantic meridional overturning circulation in FGOALS-g2. *Adv. Atmos. Sci.*, submitted.

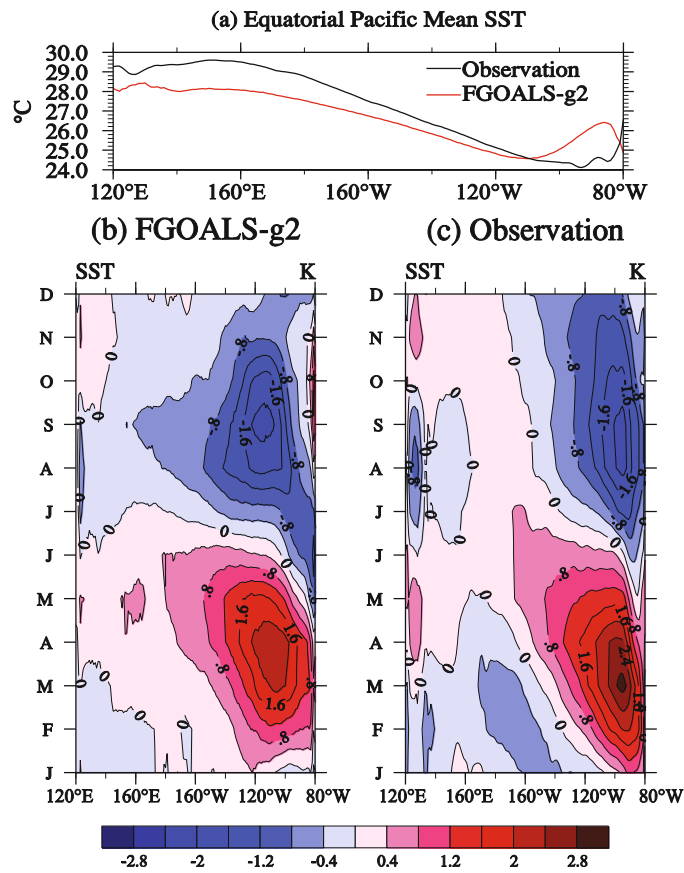


Fig. 2. (a) Mean SST along the equator in the Pacific Ocean, annual cycle of SST anomalies for (b) FGOALS-g2 20th century run, and (c) Hurrell et al. (2008) data (units: °C).

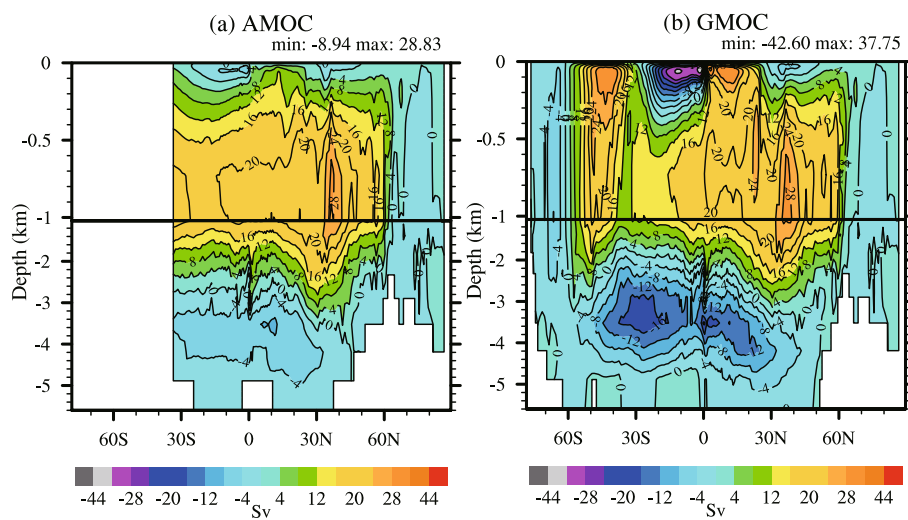


Fig. 3. (a) AMOC and (b) GMOC from FGOALS-g2 PI control run (model year 540–590, averaged) (units: Sv).

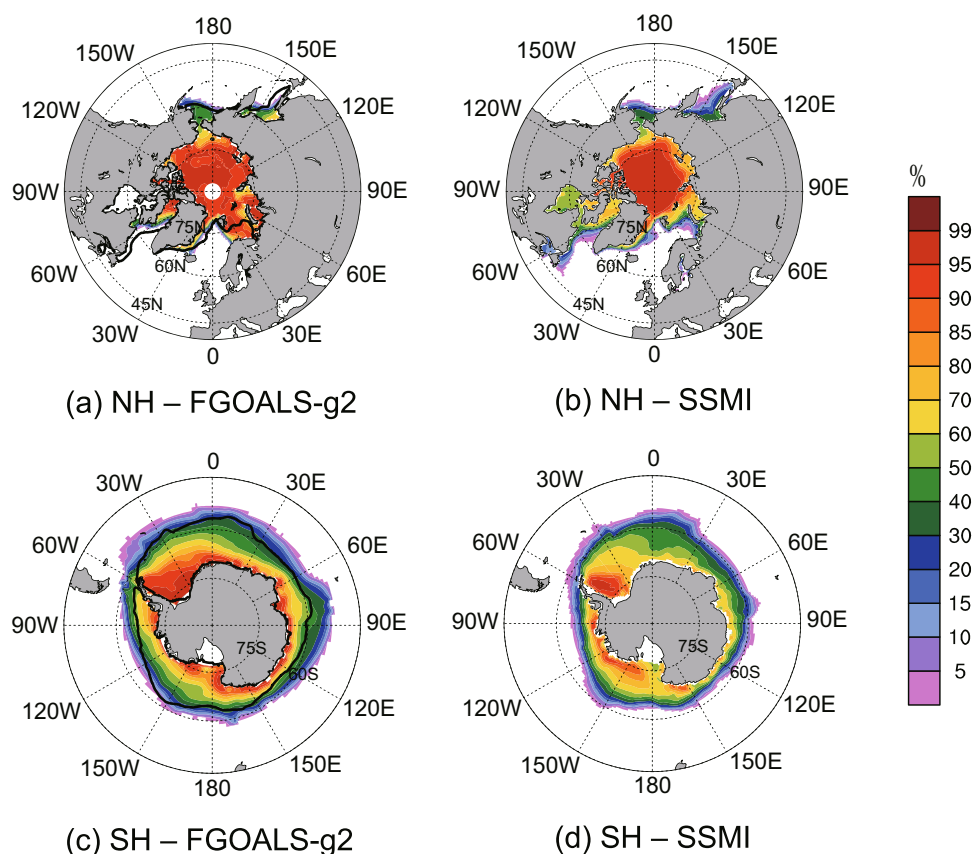


Fig. 4. (a, b) Annual mean Arctic and (c, d) Antarctic sea ice concentration (%) for FGOALS-g2 control run and SSMI observation.

less ice in the Labrador Sea and southern Sea of Okhotsk. Also, along the Eurasian continent, a higher ice concentration of $>80\%$ is produced by the model, while the observed data show a lower concentration of around 60% . For the Antarctic, FGOALS-g2 produces a relatively more extensive ice extent and a higher ice concentration as compared to observations.

Figure 5 shows spatial distributions of sea ice salinity. In the Arctic, high ice salinity is located at the sea-ice edge zones (i.e. the Bering Sea, the Sea of Okhotsk, Barents Sea, and Baffin Bay), while low ice salinity is located in the interior of the Arctic Ocean. In general, this agrees with empirical observations since the sea-ice edge zones are dominated by first-year ice with higher salinity, while the interior of the Arctic Ocean is dominated by multi-year ice with lower salinity. In the Antarctic, the relatively high ice salinity is more towards the mid-latitudes of the Southern Ocean, which is not consistent with empirical observations, since sea ice in the Antarctic is mostly formed near the coastal regions (younger ice/high salinity) and then advected equatorward (older ice/low salinity).

3.1.4 Precipitation

The frequency against daily precipitation rate over land between 20°N and 20°S is shown in Fig. 6. Results from the FGOALS-g2 20th century run are plotted with the observational estimates from the Global Precipitation Climatology Project (GPCP), which was firstly interpolated into the model grid. Except for a little overestimation of the number of light precipitation events, FGOALS-g2 produces a realistic number of moderate to very strong precipitation events. It is consistent with the performance of GAMIL2 during the Tropical Warm Pool-International Cloud Experiment (TWP-ICE) at Darwin Station with CAPT, presenting more heavily convective and stratiform precipitation events attributed to the upgraded deep convective parameterization and changes of its parameters (Li et al., 2012). The same cause-effect relationship has been pointed out in NCAR CCSM4 and Geophysical Fluid Dynamics Laboratory (GFDL) AM2 (Wilcox and Donner, 2007; Gent et al., 2011).

The double ITCZ with too much precipitation in the central Pacific near the equator and too little rain-

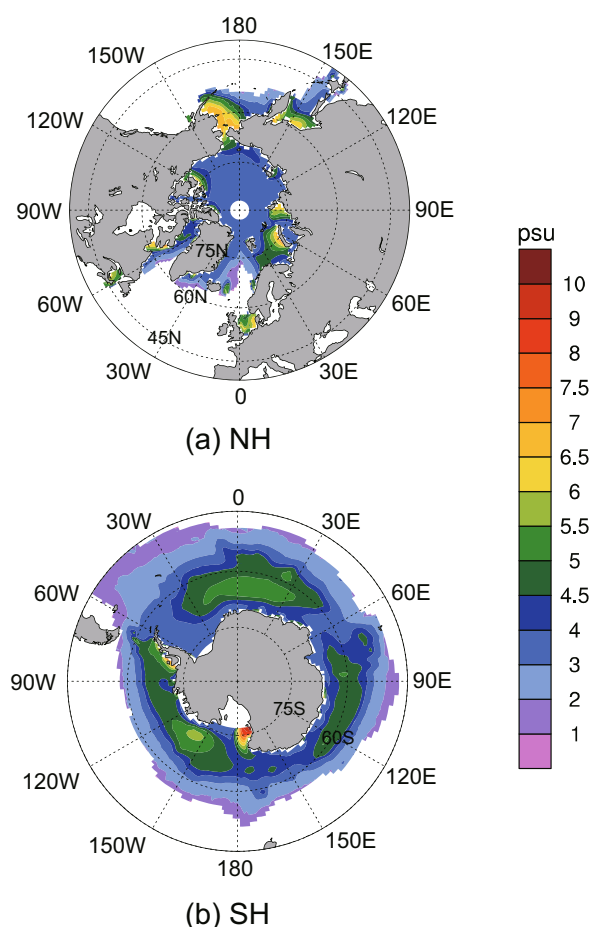


Fig. 5. (a) Annual mean Arctic and (b) Antarctic sea ice salinity for FGOALS-g2 control run (units: psu).

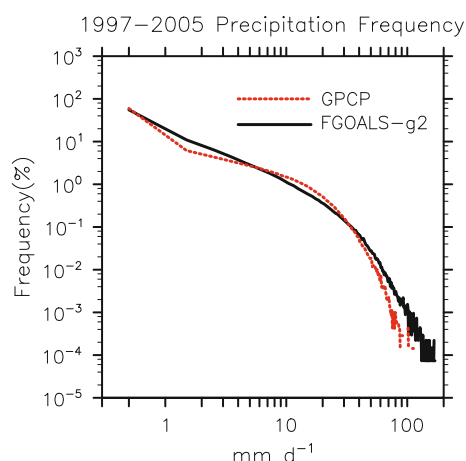


Fig. 6. Frequency of daily precipitation rate over land between 20°N and 20°S from GPCP observations (red) and 20th century runs of FGOALS-g2 (black).

fall in the west and central Pacific between 10°S and 30°S, and under-/overestimation in the Amazon Basin/Northeast Brazil and along the Andes Moun-

tains, common to this class of model (Fig. 7), are evident in both the CMIP and Atmospheric Model Intercomparison Project (AMIP) runs (Li et al., 2013). Different from the biases of other models, the two branches of the double ITCZ are especially closer to the equator, similar to its previous version (Lin, 2007). In addition, the biases in the warm pool and East Asian Monsoon region are moderately alleviated, smaller than 2 mm d^{-1} . However, the dipole bias in the tropical Indian Ocean is still large.

3.1.5 Land climatology

Figure 8 shows the annual mean land-surface temperatures from the FGOALS-g2 20th century runs compared to the observational datasets (Willmott and Matsuura, 2001). The results analyzed here are the 1950–99 historical simulations of the four-member ensemble of FGOALS-g2 20th century runs. Simulated temperature has the highest values over tropical regions, and decreases following the increasing latitude (Figs. 8a, b). The spatial distribution of the annual mean land-surface temperature is simulated well and the globally averaged bias is -0.12°C .

All large bias areas are those with higher altitude or higher latitude. It is obvious that the largest cold bias locates in the central region of the Tibetan Plateau, and this may be due to the bias of simulating the snow cover fraction (Xia et al., 2013). A higher snow cover fraction on the Tibetan Plateau leads to higher solar radiation reflectivity and less land energy absorption. In addition, the inaccurate description of topography and cryosphere in the model and extrapolation of observations in valleys over all mountain regions in Willmott and Matsuura (2001) also contribute to the bias.

Figure 9 shows the differences of total soil moisture between spring (March–April–May; MAM) and autumn (September–October–November; SON). The Global Land Data Assimilation System (GLDAS) (Syed et al., 2008) analysis is used for validation and the simulation used here is from the PI control run (541–590) of FGOALS-g2 since there is no significant qualitative difference between the historical runs and PI control runs. The spatial distribution pattern of seasonal differences in simulated soil moisture is in good agreement with GLDAS, where notable positive areas lie in Central and eastern South America, along the western coast of North America, and in South-Central Africa. Remarkable negative areas mainly lie in North-Central Africa and South Asia. However, the magnitudes of the simulation are much smaller than GLDAS, which indicate that the seasonal variation of total soil water content is too small. This may be caused by unrealistic hydrology and snow parameterization schemes in the land surface model.

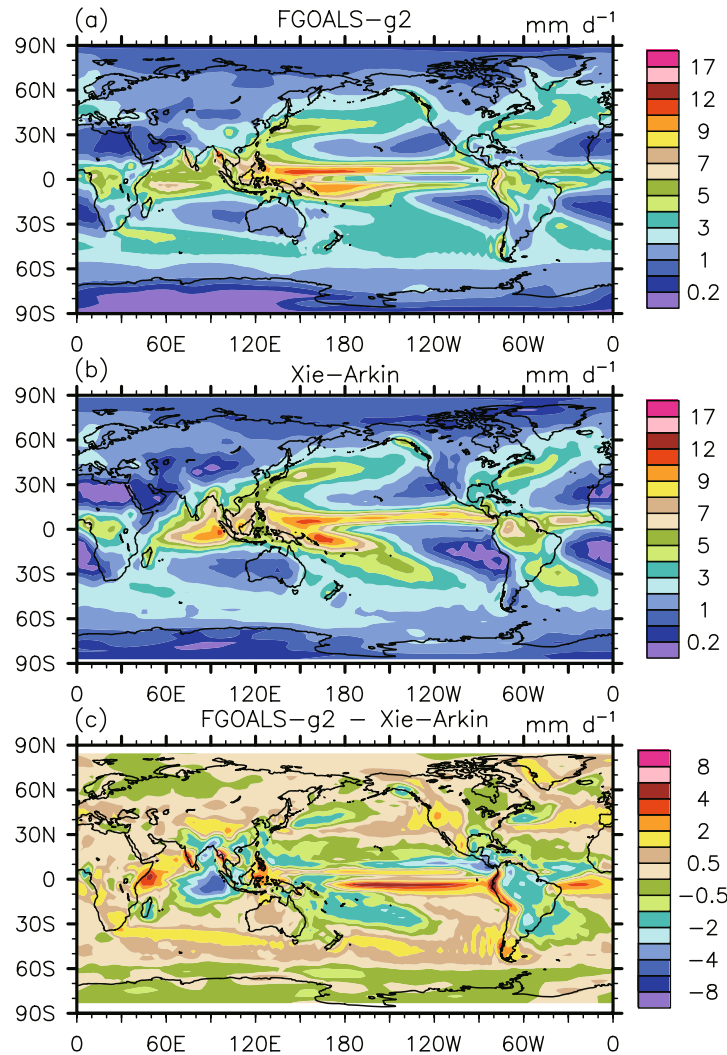


Fig. 7. Annual mean precipitation from (a) FGOALS-g2 historical runs, (b) Xie-Arkin, and (c) their differences (units: mm d^{-1}).

3.2 Climate variability

3.2.1 MJO

Following the method suggested by Waliser et al. (2009), the structure of the leading modes of combined multivariate EOF for observed 20–100-day band-pass filtered anomalies of outgoing longwave radiation (OLR) and 200-hPa and 850-hPa winds between 15°N and 15°S against longitude was calculated and is plotted in Fig. 10. In the 20th century run, FGOALS-g2 produces the correct phase relationship between equatorial convection and wind anomalies, which is sometimes misrepresented in models (e.g. Maloney and Hartmann, 2001). The key features of the MJO, such as the out-of-phase relationship between U200 and U850, the predominance of positive anomalies of U850 near and to the west of enhanced

convection (negative anomalies of OLR), the concentrated amplitude of convection anomalies in the Eastern Hemisphere, and the strong amplitude of wind anomalies across both the Eastern and Western Hemispheres, particularly at upper levels are all simulated well by FGOALS-g2, except for an eastward shift of the phase of the variables compared with the observations, coinciding with its atmospheric component performance (Xie et al., 2012). This improvement mainly results from its upgraded atmospheric physical schemes (Xie et al., 2012), especially the revised closure assumption and relative humidity threshold of the deep convection parameterization (Zhang and Mu, 2005a, b).

The principal component (PC) time series of the first two multivariate EOF modes (hereinafter EOFs)

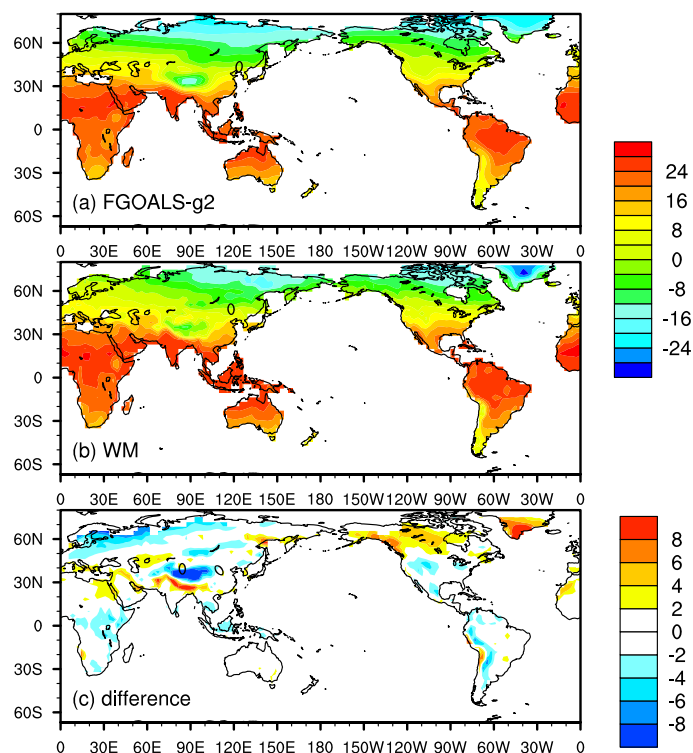


Fig. 8. Annual mean land surface temperature ($^{\circ}\text{C}$) for 1950–99 from 20th century runs: (a) simulation; (b) observations (Willmott and Matsuura, 2001); (c) difference between simulated and observed.

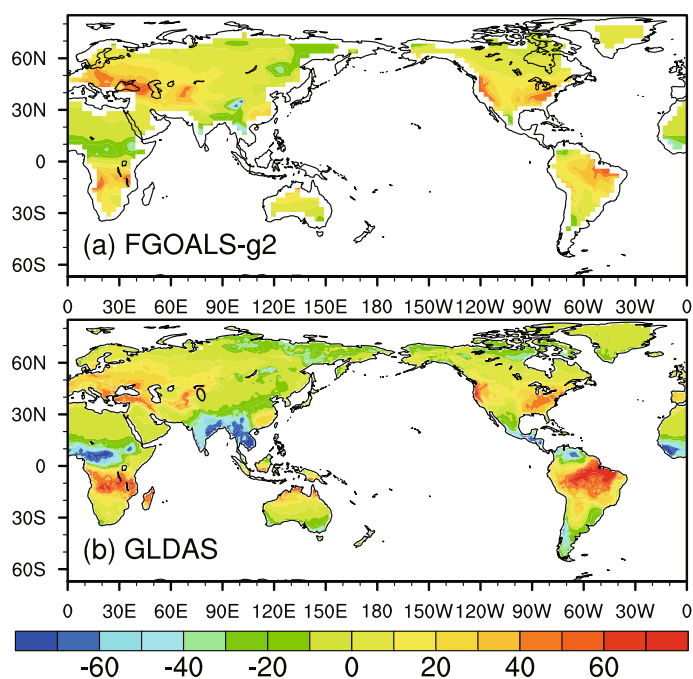


Fig. 9. Difference in mean total soil water content (kg m^{-2}) between spring (MAM) and fall (SON): (a) FGOALS-g2 control test (model years 541–590); (b) GLDAS data.

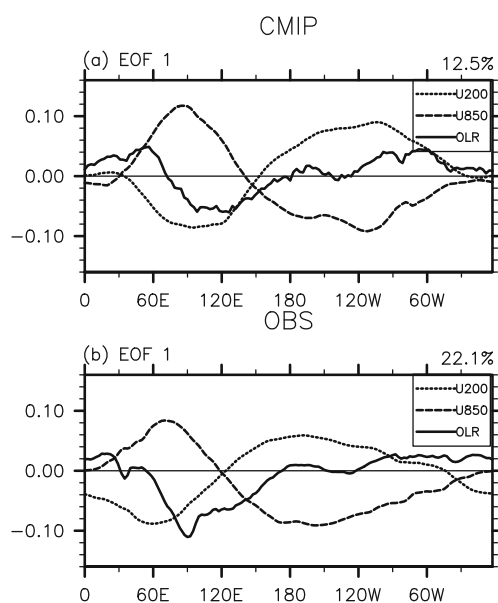


Fig. 10. Structure of the first combined multivariate combined EOF analysis of meridional mean U200, U850, and OLR from 15°S to 15°N for (a) FGOALS-g2 historical run and (b) NCEP reanalysis II/NOAA.

are correlated at a coefficient greater (or less) than 0.7 (or -0.7) when PC1 (i.e. the PC for the first mode) leads (or lags) PC2 (for the second mode) by about 10 days (or -10 days) (figure not shown). FGOALS-g2 successfully reproduces this phase relationship, which is a fundamental criterion that a simulated MJO must meet (Waliser et al., 2009) (figure not shown). The phase relationship indicates that the leading multivariate EOFs are a quadrature pair, representing coherent eastward-propagating intraseasonal variability in precipitation and winds along the equator with a period of near 40 days. The MJO index is based on the first two EOFs of the combined fields of 850-hPa and 200-hPa zonal winds obtained via the National Centers for Environmental Prediction (NCEP)–NCAR reanalysis dataset, and OLR data measured by National Oceanic and Atmospheric Administration (NOAA) polar-orbiting satellites proposed by Wheeler and Hendon (2004). The simulated 91-day running mean MJO index reproduces the tendency for the MJO to be more active during southern summer and autumn, which is reflected in the observed (figure not shown). Furthermore, the model successfully simulates some major bursts of MJO activity, such as those during early 1988 and early 1997 (figure not shown).

3.2.2 ENSO

The simulated Niño3 index exhibits prominent irregular oscillation as in the observed, but with a slightly stronger amplitude (Figs. 11a and b). The

standard deviation of Niño3 index from Hadisst is about 0.79 K, and 0.83 K from the model. Overall, it is much improved compared with the previous versions of FGOALS (Yu et al., 2008, 2011), in which the ENSO is extremely strong, e.g. the standard deviations are 2.1 K and 1.35 K for the versions g1.0 and g1.1 of FGOALS, respectively. The power spectra of the observed Niño3 index are characterized by peaks at around 4–5 yr (Figs. 11c and g), while at around 3–4 yr produced by FGOALS-g2 (Figs. 11e and h), a little longer than 3 yr by both its two versions. Some studies have argued that the air-sea interaction in the Indian Ocean might have an important influence on the simulated ENSO period (Yu et al., 2002), and thus it is necessary to explore the detailed relationship between air-sea interaction processes in the tropical Indian and Pacific Oceans in the future. The phase-lock is another important phenomenon in the observed ENSO, which implies a key role of the seasonal cycle in ENSO dynamics. As first indicated by Rasmusson and Carpenter (1982), SST anomalies associated with ENSO events begin to develop in boreal spring, and then reach a maximum in boreal winter, before finally diminishing the next spring (Fig. 11d). The phase-lock characteristics of the ENSO cycle can be reproduced well by the coupled GCM (Fig. 11f).

3.2.3 North Atlantic Oscillation

Figure 12 shows the annual mean December–January–February (DJF) sea level pressure (SLP) pattern and the leading EOF mode derived from the historical run of FGOALS-g2 and NCEP2 reanalysis. The simulated SLP pattern in the Atlantic sector of the Northern Hemisphere is similar to the well documented observational pictures (Zhou et al., 2000); only the Azores High and the Icelandic Low are weaker than the observations. The North Atlantic Oscillation (NAO) pattern is identified by carrying out EOF analysis. As shown in Figs. 12c and d, the well known meridional dipole structure is reasonably simulated, except that the subtropical center shifts westward relative to the reanalysis. The intensity of the subtropical center is also weaker than the reanalysis (Zhou et al., 2000). A previous analysis on FGOALS-g1 found that the model shows moderate skill in reproducing the Arctic Oscillation (AO)/NAO pattern, and the simulated subtropical Atlantic sector of the NAO pattern is also weaker than the reanalysis (Xin et al., 2008). In addition, we also compared the interannual standard deviation of SLP in the simulation with the reanalysis, and found that the simulation is weaker than the reanalysis, in particular in the Icelandic and Barents Sea sectors where the air–ice interaction is very active

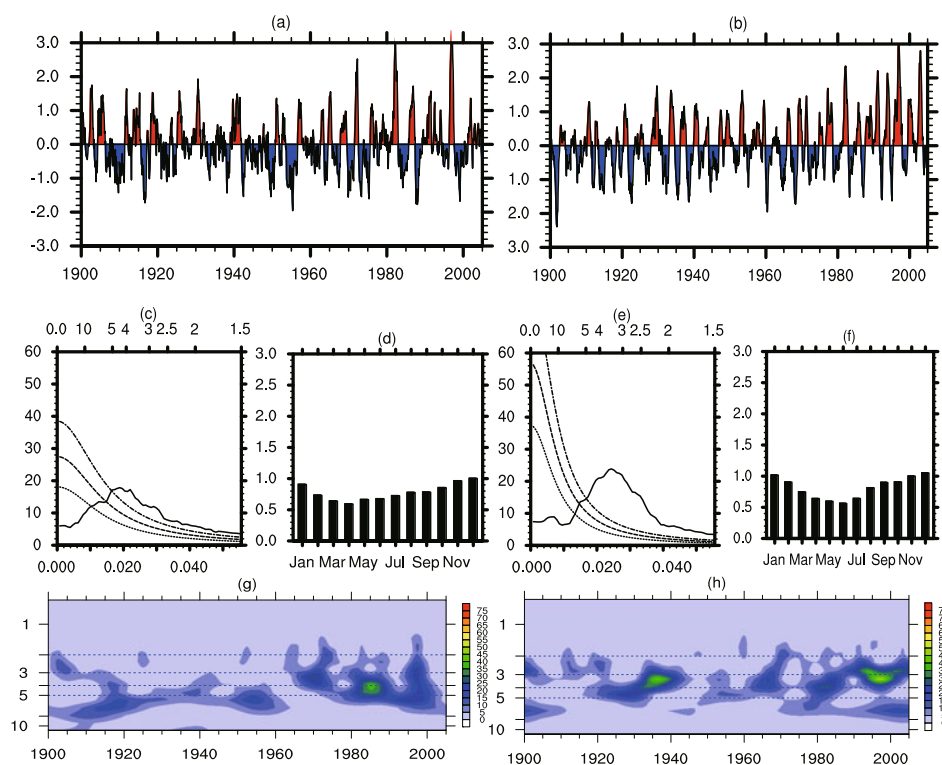


Fig. 11. (a, b) Niño 3 index, and (c, e) its power spectrum, (d, f) standard deviation for each calendar month and (g, h) Morlet wavelet analysis for HadISST (left) and FGOALS-g2 historical run (right).

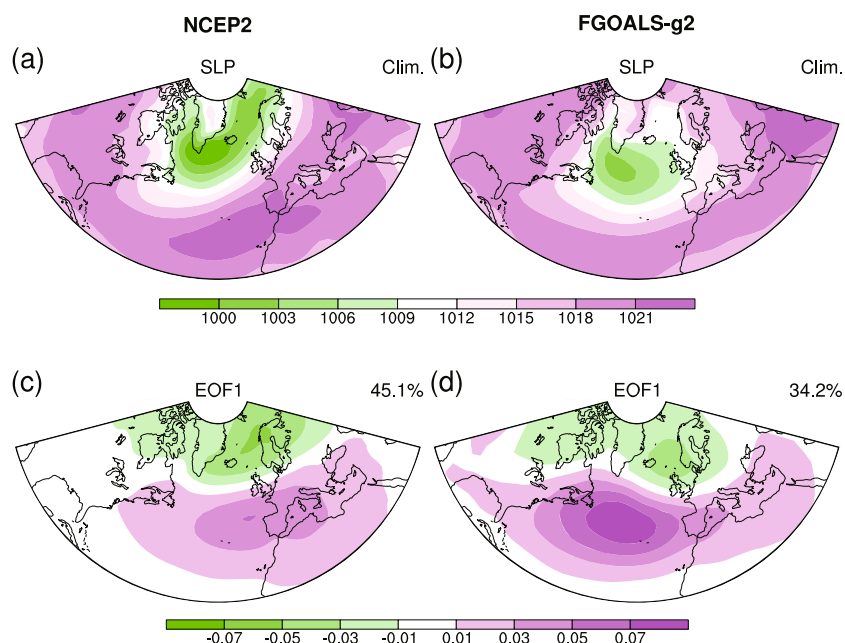


Fig. 12. DJF averages of (a, b in hPa) sea level pressure and (c, d) the first EOF mode of DJF SLP covariance matrices. Figures 14a, c are based on FGOALS-g2 historical run, and 14b, d on NCEP2 reanalysis data for the period 1979–2009.

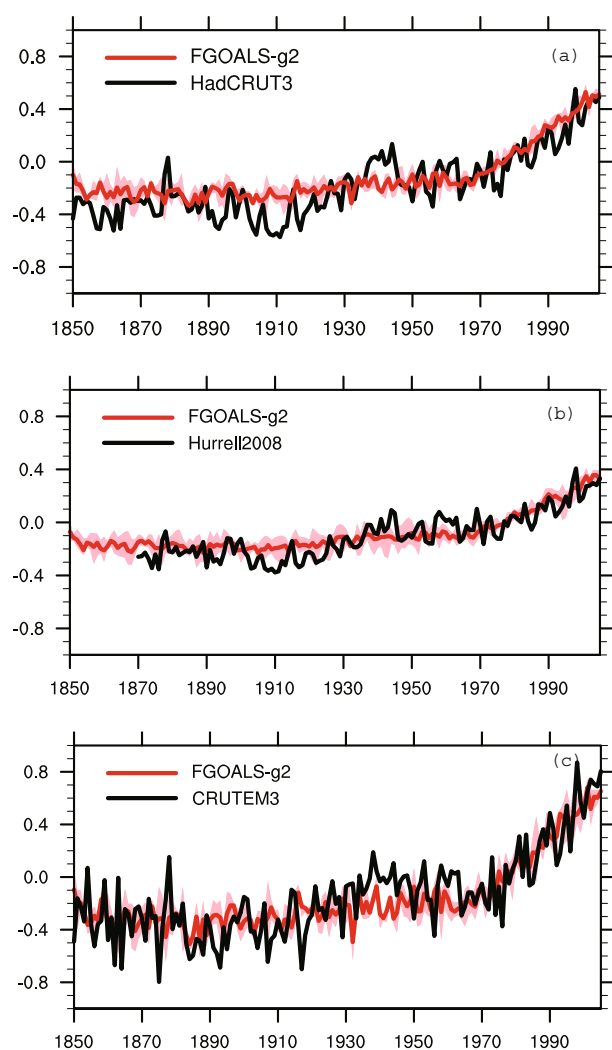


Fig. 13. Time series of (a) global, (b) ocean and (c) land average surface temperature anomaly ($^{\circ}\text{C}$) from observation (black) and FGOALS-g2 20th century ensemble mean (red). The shading shows the maximum and minimum of the FOGALS-g2 ensemble spread.

(figure not shown).

3.3 Evolution of 20th century runs

Figure 13 shows the time series of global, ocean and land average surface temperature anomaly from observations and the ensemble mean from four 20th century runs of FGOALS-g2. The observations indicate that the surface warmed from 1910–40, cooled slightly from 1940–70, and then warmed markedly from 1970 onward (IPCC, 2007a; Thompson et al., 2008). The model reproduces the globally averaged annual mean surface temperature evolution in the past 156 years quite well, especially the recent notable warming of both land and ocean, but also the slight warming and cooling of land from 1910–40 and 1940–70. However,

it simulates a smooth SST evolution before 1970 and small inter-annual SST variation compared with the observation. The SST ensemble spread is also smaller than that of land surface temperature.

Figure 14 shows the latitude-time sections of zonal mean land surface temperature anomalies ($^{\circ}\text{C}$) from 1900–2010, relative to the 1961–90 mean. Climate Research Unit (CRU) (Brohan et al., 2006) temperature data were used to validate model results. Generally speaking, the simulated results of temporal variation of zonal mean temperature anomalies, which are lower than the 1961–90 average before the 1980s and higher after the 1980s, are much similar to the observation, as respectively shown in green and red in Fig. 14. The Northern Hemisphere has higher anomalies compared to the Southern Hemisphere after the 1980s and lower anomalies before the 1980s for both simulated and observed results. In spite of that, the interdecadal variability in simulated temperature is smaller than the observation.

3.4 Asian and African Monsoon

Asian and African Monsoon precipitation and circulation in boreal summer are shown in Fig. 15. In the Asian Monsoon, the five major precipitation centers in the tropics, such as the southern rim of India, Bay of Bengal, South China Sea, Philippine Sea and the southeastern tropical Indian Ocean, are reproduced by the model, while they are far more intensive over India and the Bay of Bengal, much weaker over the southeastern tropical Indian Ocean, and slightly weaker over the western northern Pacific than the observation. The rainband in the subtropics extending from the middle and lower reaches of the Yangtze River to the east of Japan simulated by FGOALS-g2 is somewhat weak and shifting eastward out of the Asian continent. The discrepancy may be associated with the strong eastward wind and weak western Pacific subtropical high in the model. Although the spurious rainfall still exists in the southeast of the Tibetan Plateau, it is largely alleviated compared to its previous version.

In the African Monsoon, the model captures the two precipitation maxima along the coast of West Africa and the Gulf of Guinea, respectively, and the northern edge of the summer monsoon precipitation in Northeast Africa is consistent with the CPC Merged Analysis of Precipitation (CMAP) data. The total precipitation is slightly larger than the observation, which is related with a stronger convection within the ICTZ around 10°N (figure not shown). The West African heat low is also captured by the model; however, the convection reaches too high in altitude and also expands northward towards the Saharan region (figure not shown), which leads to the excess precip-

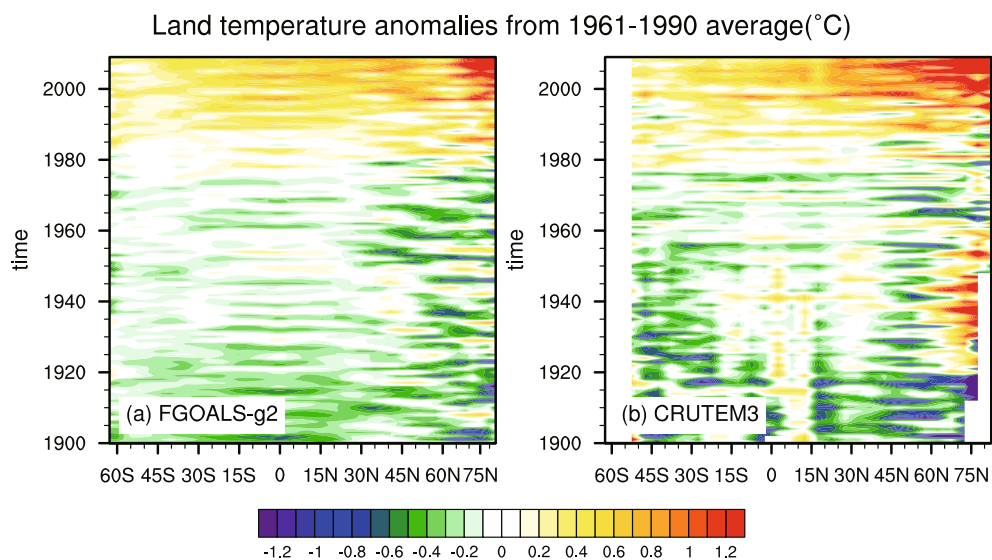


Fig. 14. Latitude-time sections of zonal mean temperature anomalies of land surface from 1900–2005, relative to the 1961–99 mean: (a) FGOALS-g2 historical runs; (b) observation (CRUTEM3) (units: °C).

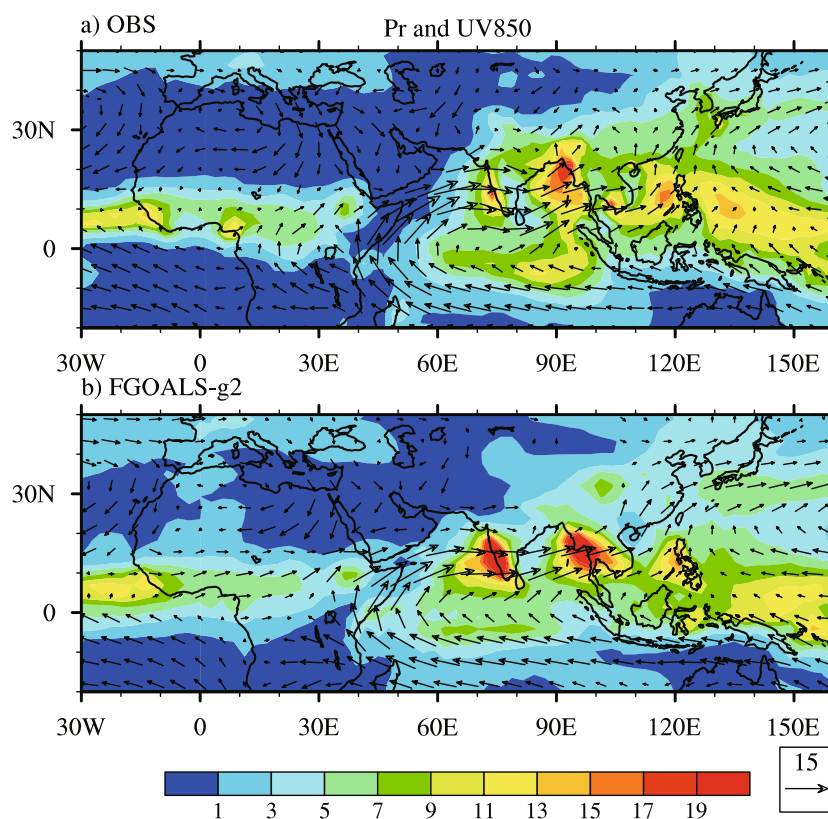


Fig. 15. JJA-mean climatological precipitation and 850-hPa wind for (a) ERA-40/CMAP and (b) FGOALS-g2 PI run.

itation over Northwest Africa, which is not shown in the observation.

4. Summary and discussion

FGOALS-g2, one of the newest versions of FGOALS, and its basic performance in the PI control run and 20th century runs have been investigated in this study. The results indicate that there are many fundamental improvements without incurring any known new biases, compared with those from FGOALS-g1. First is the annual cycle of SST along the equator in the Pacific (Fig. 2b), which could be attributed to the higher resolutions (0.5° meridional resolution in the tropics and 10 m for each layer in the upper 150 m), the updated mixing scheme in the ocean component, and the improvement of the cumulus convection scheme in the atmospheric component. Second is the ENSO simulation, including its irregular period by peaks around 3–4 years, the amplitude with a standard deviation of about 0.83 K, and the phase-lock. The improvements in the amplitude and period correspond with the improved latitudinal width of the anomalous equatorial zonal wind stress—about 20° for the 300-yr control run—well within the observational 19° – 24° ranges (Capotondi et al., 2006; Deser et al., 2012). The better simulations of the frequency of tropical land precipitation, East Asian Monsoon precipitation and its seasonal cycle, MJO and ENSO are closely related to the updated cumulus parameterization and the alleviation of uncertainties in some key parameters in shallow and deep convection schemes, cloud fraction, cloud macro/microphysical processes, and boundary layer scheme in GAMIL2. The influence of these parameters with uncertainty in the model is a hot topic, which will be further discussed in other studies. The 20th century surface temperature increase by FGOALS-g2 is close to the observation, especially the notable recent warming after the 1970s, which probably results from the inclusion of aerosol indirect effects (Li et al., 2013). The spatial pattern of land surface temperature, soil water content changes between boreal spring and fall, and sea ice concentration are also captured well by FGOALS-g2, but for regional biases and small amplitudes. In addition, the sea ice salinity simulation is one special characteristic of FGOALS-g2, although it is inconsistent with empirical observations in the Antarctic (Fig. 5).

However, there still exist significant biases in FGOALS-g2 simulations that need to be further studied and improved. For example, the “double ITCZ” phenomenon in both the atmosphere and ocean simulations is not well understood in FGOALS-g2, though it is a common problem in GCMs. The greater Arctic

Sea ice concentration and the biases in the Labrador Sea are probably due to the absence of the arctic penetration current in the ocean model for the former and the weaker Icelandic Low for the latter (figure not shown). In the $1^\circ \times 1^\circ$ GAMIL version, the Icelandic Low bias is remarkably alleviated (unpublished), which may be another clue to the deficiency of NAO simulation by the coupled GCM with a low-resolution atmospheric component. The spurious rainfall in the southeastern Tibetan Plateau is another problem, in spite of a certain level of alleviation in FGOALS-g2 simulations. Besides, the runoff from land to ocean is not turned on because of some technical interpolation problems, which will influence ocean salinity, as well as the AMOC.

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